

1. Palindrome

QUESTIONS 22 / 48 completed

Palindrome ▾

PALINDROME

Write a program to read in an integer and determine if it is a palindrome or not using a while loop.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     int number,original,digit,reverse=0;
5
6     cout<<"enter the number:"<<endl;
7     cin>>number;
8
9     original=number;
10
11     for(;number>0; number=number/10){
12         digit=number%10;
13         reverse = reverse*10+digit;
14     }
15     if (original==reverse)
16         cout<<"it is palindrome"<<original;
17     else
18         cout<<"it is not palindrome";
19     return 0;
20 }
```

INPUT

121

2.GCD

QUESTIONS 22 / 48 completed

GCD

GCD

Write a program to read in two integers and compute their greatest common divisor using a do-while loop.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     int a,b,remainder;
5
6     cout<<"enter two number";
7     cin>>a>>b;
8
9     if (b==0){
10         cout<<"gcd is:"<<a;
11         return 0;
12     }
13     do{
14         remainder=a%b;
15         a=b;
16         b=remainder;
17     }while(b!=0);
18     cout<<"the gcd is :"<<a;
19     return 0;
20 }
```

INPUT

2 5

3Happy

QUESTIONS 23 / 48 completed

Happy

HAPPY

Write a program to read in an integer and determine if it is a happy number or not.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int num, sum, rem;
6     cin >> num;
7
8     while (num != 1 && num != 4) {
9         sum = 0;
10        while (num > 0) {
11            rem = num % 10;
12            sum += rem * rem;
13            num /= 10;
14        }
15        num = sum;
16    }
17
18    if (num == 1)
19        cout << "Happy Number";
20    else
21        cout << "Not Happy Number";
```

INPUT

120

4.Armstrong

QUESTIONS 22 / 48 completed

Armstrong

ARMSTRONG

Write a program to read in an integer and determine if it is an Armstrong number or not.

PROGRAM EDITOR

C++

Run Save

```
1 #include <iostream>
2 #include <cmath>
3 using namespace std;
4 int main() {
5     int num, originalNum, remainder, digits = 0;
6     double sum = 0.0;
7     cout << "Enter an integer: ";
8     cin >> num;
9     originalNum = num;
10    int temp = num;
11    while (temp != 0) {
12        temp /= 10;
13        digits++;
14    }
15    temp = num;
16    while (temp != 0) {
17        remainder = temp % 10;
18        sum += pow(remainder, digits);
19        temp /= 10;
20    }
21    if ((int)sum == originalNum)
```

INPUT

123

5Hrashad

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     int num,temp,sum=0;
5     cin>>num;
6     temp=num;
7     while(temp>0){
8         sum+=temp%10;
9         temp/=10;
10    }
11    if(num%sum==0)
12        cout<<"Harshad Number";
13    else
14        cout<<"Not Harshad Number";
15    return 0;
16 }
```

INPUT

555